

CO2 AT 1

46. $S \rightarrow aA \mid b$

$A \rightarrow dA \mid \epsilon$

input a d d

$S \rightarrow aA$

$S \rightarrow b$

$A \rightarrow dA$

$A \rightarrow \epsilon$



$FIRST(S) = \{a, b\}$ $FOLLOW(S) = \{\$ \}$

$FIRST(A) = \{d, \epsilon\}$ $FOLLOW(A) = \{a, b, d, \epsilon, \$ \}$

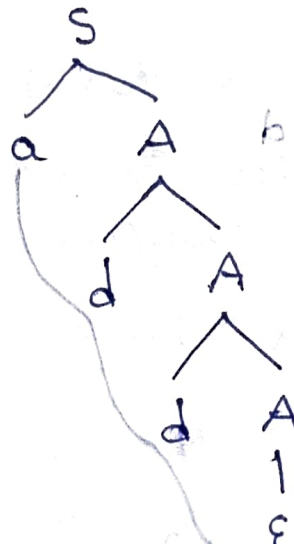
predictive parser table:

Non-Terminal	a	b	d	ϵ
S	$S \rightarrow aA$	$S \rightarrow b$		
A			$A \rightarrow dA$	$A \rightarrow \epsilon$

stack table:

stack	input	output
$\$ S$	a d d $\$$	
$\$ A a$	a d d $\$$	$S \rightarrow aA$
$\$ A d$	d d $\$$	$A \rightarrow dA$
$\$ A d$	d $\$$	$A \rightarrow dA$
$\$$	$\$$	$A \rightarrow \epsilon$

parser tree:



d | a b c - A
 3 | a b c - A
 b b a - a
 a b c - a
 d a - c
 a b c - a
 3 c - a

{P} = {a} was 07 {d} = {a} was 17
 {a} = {a} was 07 {3, b} = {a} was 17
 = a d d

3/d/d/d was 07 was 17 was 17

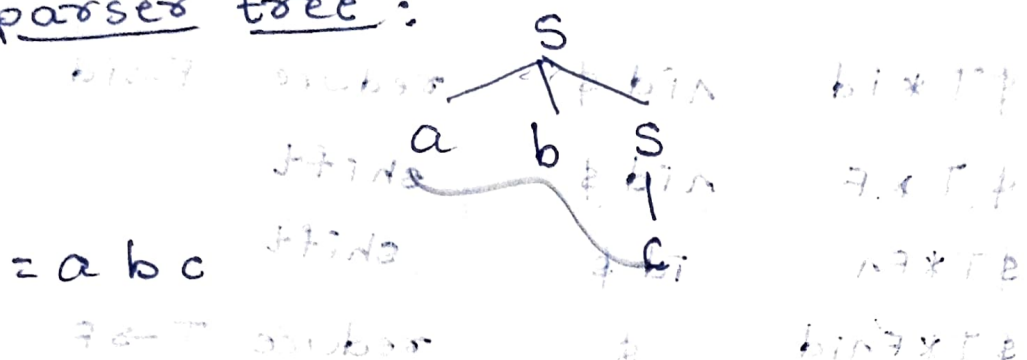
47. $S \rightarrow abS \mid c$
 input : abc

$S \rightarrow abS$

$S \rightarrow c$

stack	input	action
\$ S	abc \$	
\$ Sba	abc \$	$S \rightarrow abS$
\$ Sb	bc \$	
\$ c	c \$	$S \rightarrow c$
\$	\$	accept

parser tree:



48.

Given expression,

$id \wedge id * id \wedge id$

Grammar: $E \rightarrow E * T \mid T \quad T * T \mid$

$T \rightarrow F \wedge T \mid F T * \exists$

$F \rightarrow id$

$\wedge$ has highest precedence than $*$.

stack	input	action
\$	id id * id id	shift
\$ id	id * id id	reduce
\$ F	id * id id	$F \rightarrow id$ shift
\$ F \wedge	id * id id	shift
\$ F \wedge id	* id id	reduce $F \rightarrow id$
\$ F \wedge F	* id id	reduce $T \rightarrow F$
\$ F \wedge T	* id id	reduce $T \rightarrow F \wedge T$
\$ T	* id id	shift
\$ T *	id id	shift

\$T * id	id \$	reduce $F \rightarrow id$
\$T * F	id \$	shift
\$T * F \wedge	id \$	shift
\$T * F id	\$	reduce $T \rightarrow F$
\$T * T id	\$	reduce $F \rightarrow id$
\$T * T \wedge F	\$	reduce $T \rightarrow T \wedge F$
\$T * T	\$	reduce $E \rightarrow T$
\$E * T	\$	reduce $E \rightarrow E * T$
\$E	\$	accept

parse tree:

